

SSD_Emmc usage reference

1 Overview

At present, SSD20X can support Emmc in Uboot/Kernel stage, that is to say, it supports booting the Rootfs/Kernel partition from Emmc after starting from Uboot

Common usage scenarios: Nor (small capacity) + Emmc startup, the partition before (including) Uboot is placed on Nor Flash, and the partition after Uboot is placed on Emmc for use

SSD20X has only one SDIO, so to use EMMC, you need to close SDMMC and then open EMMC

2. Uboot stage configuration

Close SDMMC CONFIG:

```
#CONFIG_MS_SDMMC is not set
```

Open EMMC CONFIG:

```
CONFIG_MS_EMMC=y  
CONFIG_SUPPORT_EMMC_BOOT=y
```

As shown below, first close MSTAR SDMMC through make menuconfig, then open MSTAR eMMC and Support eMMC BOOT

```
make menuconfig
#Device Driver
  [*]Mstar driver
    [*]MSTAR emmc
    [*]support eMMC BOOT
    [ ]MSTAR sdmmc (do not check)
```

3. Kernel stage configuration

a. Turn off SDMMC and enable EMMC config support:

```
CONFIG_MS_EMMC=y
CONFIG_MS_EMMC_UNIFY_DRIVER=y
# CONFIG_MS_SDMMC is not set
```

b. Open the file system to support EMMC partitions:

```
CONFIG_PARTITION_ADVANCED=y
CONFIG_EMMC_PARTITION=y
```

c. Open the ext4 related config:

```
CONFIG_LBDAF = y (support ext4 large-capacity block devices and
files)
CONFIG_EXT4_FS = y (open support for ext4 file system ,
CONFIG_EXT4_FS sub-option config can be kept by default )
```

Suboptions after ext4 config is opened :

```
CONFIG_EXT4_USE_FOR_EXT2=y
CONFIG_JBD2=y
CONFIG_FS_MBCACHE=y
CONFIG_CRYPTOCRC32C=y
```

参照下图，使用make menuconfig去配置：

```

make menuconfig
#Enable the block layer
[*]Support for large (2TB+) block devices and files ( emmc启动时需要 )
[*]Partition Types
    [*]EMMC PARTITION table support
#Device Drivers
    [*]MMC/SD/SDIO card support
        [*]HW reset support for eMMC
    [*]Simple HW reset support for MMC
    [*]MMC block device driver
[*]SStar Soc platform drivers
    [*]EMMC driver
    [*]UNIFY EMMC DRIVER
    [*]MMC block device driver
    [ ]SStar SD/MMC Card Interface Suport ( 不要选中 )
#File systems
    [*]The Extended 4 (ext4) filesystem ( emmc启动时需要 · 子选项保持默认 )

```

4. 公版已经做好的EMMC分区配置参考

因SSD20X不支持空片从EMMC启动，Uboot之前的必须从SPI FLASH启动，所以EMMC分区需要配合Nand/Nor Flash组合使用，常见使用是小容量Nor+EMMC

	Nand + EMMC	Nor + EMMC
project	./setup_config.sh ./configs/nvr/i2m/8.2.1/emmc_spinand.glibc-ext4fs.011a.64	./setup_config.sh ./configs/nvr/i2m/8.2.1/e ext4fs.011a.64
kernel	make infinity2m_emmc_spinand_ssc011a_s01a_defconfig	make infinity2m_emmc_nor_ssc
uboot	make infinity2m_spinand_emmc_defconfig	make infinity2m_nor_emn